Architectural Morphogenesis in Autonomous Agent Systems

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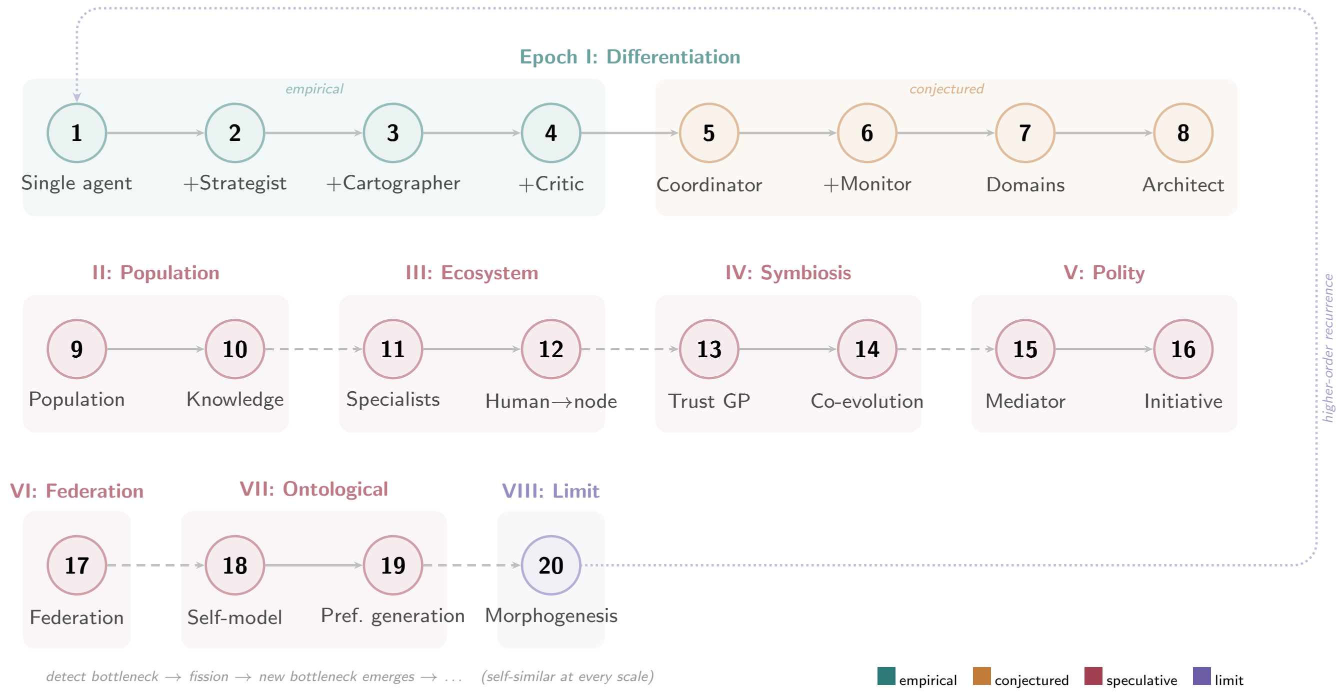

Figure 1: Twenty phases of architectural morphogenesis across eight epochs, graded by epistemic status: observed in the Wallfacer experiment (teal), conjectured from the fission principle (orange), speculatively extended to populations and self-reference (red), and the limiting claim that the stable attractor is morphogenesis itself (purple). Each transition is driven by bottleneck migration: resolving one constraint reliably exposes the next. The dotted loop from Phase 20 back to Phase 1 marks the claim that the entire sequence recurs at higher orders of abstraction.

ABSTRACT

This paper develops a theoretical framework for understanding the architectural evolution of autonomous agentic systems. Drawing on a longitudinal case study in which an LLM-based agent pipeline operated without human intervention for seven days, we identify a recurring phenomenon we term *bottleneck migration*: resolving a performance constraint in one dimension reliably relocates it to another, rather than producing net improvement. We propose a *fission principle* that accounts for the observed pattern of role differentiation in multi-agent architectures, and we formalize the trust relationship between agents and human overseers as a Gaussian Process preference learning problem. The framework extends, in a more speculative register, to populations of agent instances, multi-principal preference aggregation, and co-evolutionary dynamics. We examine the ontological limits of self-reconfiguring architectures through the lens of Lawvere's fixed-point theorem and Gödel's incompleteness, concluding that the stable attractor of such systems is not a particular topology but rather the sustained capacity for topological change. Throughout, we are explicit about which claims rest on empirical observation and which constitute theoretical conjecture intended to guide future investigation.

KEYWORDS

agentic AI, autonomous systems, preference learning, Bayesian optimization, self-reference, organizational morphogenesis, human-in-the-loop

1 INTRODUCTION

The verification regress, wherein a verifier employed to verify a claim itself requires verification, constitutes one of the oldest problems in epistemology. Known variously as Agrippa's trilemma [46] or Münchhausen's trilemma [1], it admits three classical responses:

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infinite regress, foundationalism, and coherentism. With the emergence of autonomous agentic AI systems, that is, pipelines of LLM-based agents that plan, execute, document, and verify their own work without continuous human supervision, this philosophical puzzle has acquired practical engineering significance. When an LLM agent generates code and another LLM agent reviews it, neither agent's judgment can serve as an ultimate ground of correctness. The question of how to structure such systems so that they remain reliable despite the absence of a privileged verifier is no longer purely academic.

This paper argues that the verification regress, rather than constituting merely an obstacle, functions as a generative engine of architectural differentiation in agentic systems. We ground this argument in a longitudinal case study, the Wallfacer experiment, in which an autonomous pipeline operated for seven consecutive days, and we observe that successive attempts to resolve verification failures at one level produced new architectural roles, which in turn introduced new forms of the regress at higher levels. The central empirical finding is what we term *bottleneck migration*: the phenomenon whereby improving system performance along one dimension reliably shifts the binding constraint to a different dimension.

The paper makes six contributions, which we present in decreasing order of empirical grounding. First, we report the bottleneck migration phenomenon and connect it to Goldratt's Theory of Constraints [22], Wolpert and Macready's No Free Lunch theorems [53], and Ashby's Law of Requisite Variety [6]. Second, we report a replication experiment in which three independent agent instances, started from empty repositories with no seed task, converged on building the same class of software (cellular automaton simulators) yet diverged in architecture, language, and feature focus, providing early empirical evidence for convergent evolution and adaptive radiation in agentic populations. Third, we articulate a fission principle that describes when and why role differentiation occurs, drawing on Simon's near-decomposability [48] and coordination theory [35]. Fourth, we formalize trust calibration as Gaussian Process preference learning [11, 23]. Fifth, we extend the framework, now in a more speculative mode, to populations of agent instances and multi-principal scenarios, drawing on evolutionary dynamics [9, 27] and social choice theory [5]. Sixth, and most speculatively, we examine the ontological limits of self-reconfiguring architectures through Lawvere's fixed-point theorem [33] and Gödel's incompleteness [21].

We wish to be transparent about the epistemic status of these contributions. The first two rest on direct observation from the Wallfacer and Ralph experiments, though both use the same LLM substrate and the Ralph experiment partially addresses the single-case-study limitation through replication. The third rests on observed fission events in Wallfacer, supported by negative evidence from Ralph. The fourth is a formal construction whose empirical adequacy remains to be tested. The fifth and sixth are theoretical extrapolations intended to map the conceptual space and identify productive directions for future research, not to report established findings.

2 THE WALLFACER EXPERIMENT

2.1 Motivation and Setup

The experiment was motivated by a straightforward question: what happens when an LLM-based agent pipeline is left to operate autonomously for an extended period? While single-turn and short-session evaluations of LLM agents are common in the literature, sustained autonomous operation over days or weeks remains underexplored. We sought to observe, rather than prescribe, the architectural pressures that emerge at scale.

The Wallfacer pipeline was deployed on a virtual private server running Ubuntu, with Claude Code [3] as the base LLM substrate. The pipeline was configured for a software engineering task: maintaining and extending a moderately complex codebase. It operated in a self-directed loop encompassing ideation (selecting the next task), implementation (writing code), testing (running the test suite), documentation (updating project records), and deployment (committing and pushing changes). No human reviewed or intervened in any of these stages for the duration of the experiment.

We conducted the experiment across four successive architectural configurations, each introduced when the preceding configuration exhibited sustained performance degradation that we judged, in retrospect, to stem from a structural rather than parametric limitation. The four configurations were: (1) a single agent performing all functions in a self-loop; (2) a Strategist agent separated from an Executor agent; (3) the addition of a Cartographer agent; and (4) the further addition of a Critic agent.[1] Each configuration was run until we observed a clear and persistent bottleneck, at which point we introduced the next configuration. The total duration across all configurations was seven days, during which the codebase grew from approximately 10,000 to over 40,000 lines of code.

2.2 Observations and Analysis

We report three principal observations from the experiment, along with our interpretation of each. We acknowledge that as a single case study, these observations are suggestive rather than conclusive, and that alternative interpretations are possible.

Observation 1: Bottleneck migration. Across all four configurations, resolving a performance bottleneck in one dimension of system behavior reliably relocated the binding constraint to a different dimension, rather than producing an overall improvement that persisted. For example, in the two-agent configuration, separating planning from implementation eliminated the context-competition bottleneck observed in the single-agent configuration, but introduced a new bottleneck: implicit knowledge generated during implementation (such as the rationale behind design decisions) was not captured in any durable form and therefore decayed between planning cycles. The Cartographer was introduced to address this decay, which it did, but at the cost of introducing a quality-control bottleneck: three agents were producing outputs, but none was tasked with evaluating whether those outputs met any standard of correctness.

[1] We name these roles after their cognitive mode rather than their mechanical function. The Strategist decides *what* to do (not merely "proposes"), the Executor implements (not merely "builds"), the Cartographer produces legible abstractions of system state (not merely "documents"), and the Critic exercises evaluative judgment (not merely "verifies"). These names reflect the fission principle's emphasis on competing cognitive modes.

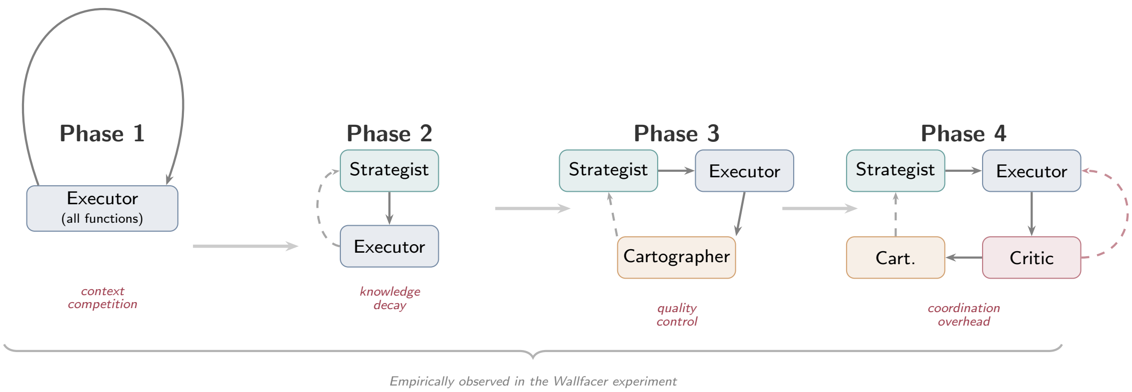

Figure 2: Architectural evolution across four observed phases. Each transition was triggered by a persistent bottleneck (shown in red). Solid arrows indicate primary information flow; dashed arrows indicate feedback.

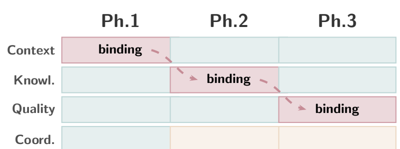

Figure 3: Bottleneck migration across phases. Dark cells mark the binding constraint; light cells indicate resolved dimensions. Dashed arrows show how each fission relocates the bottleneck.

This pattern, in which resolving one constraint exposes another, aligns with Goldratt's Theory of Constraints [22], which posits that any system has exactly one binding constraint at a time and that improving non-constraints yields no net throughput gain. It also resonates with the No Free Lunch theorems [53], which establish that no single optimization strategy dominates across all problem instances. In our setting, the analogous claim is that no single architectural configuration dominates across all bottleneck types. We note, however, that our evidence for this claim is limited to a single system operating on a single class of tasks, and that broader empirical validation is required before any general principle can be asserted with confidence.

Observation 2: Emergent self-diagnosis. In the three-agent configuration (Strategist, Executor, Cartographer), the system proposed refactoring its own monolithic codebase at approximately 51,000 lines of code. This behavior did not emerge in the two-agent configuration at a comparable scale. We interpret this as evidence that the Cartographer role, by producing structured representations of the system's architecture and history, provided the Strategist with a legible abstraction of global system state that was otherwise unavailable. In the two-agent configuration, the Strategist's decisions were conditioned only on the current codebase and task queue, without any higher-level summary of architectural trends.

This interpretation finds support in Clark and Chalmers' Extended Mind hypothesis [12], which argues that cognitive processes are not confined to the brain but extend to external artifacts that participate in reasoning. In our case, the documentation artifacts produced by the Cartographer function as external cognitive scaffolding for the Strategist: they do not merely record what has happened but actively shape what the Strategist can perceive and reason about. We are cautious, however, about overinterpreting a single instance of emergent behavior. It is possible that the refactoring proposal would have emerged eventually in the two-agent configuration at a higher line count, or that it was triggered by idiosyncratic features of the codebase rather than by the architectural difference we hypothesize.

Observation 3: Coordination cost scaling. As the number of agent roles increased from one to four, we observed a nonlinear increase in the volume of inter-agent communication required to maintain coherent system behavior. The four-agent configuration exhibited episodes of circular dependency, in which the Critic rejected output that the Executor had produced according to the Strategist's specification, leading to repeated revision cycles that consumed computational resources without advancing the project. This observation is consistent with Brooks' [10] well-known finding that communication overhead in human teams scales quadratically with team size, and it suggests that the benefits of role differentiation are subject to diminishing returns as coordination costs accumulate. Figure 4 traces the estimated trajectory of six system metrics across all eight phases, illustrating how capability saturates while coordination cost, self-awareness, and autonomy follow distinct growth curves.

3 THE RALPH EXPERIMENT: CONVERGENT EVOLUTION WITHOUT SPECIFICATION

The Wallfacer experiment demonstrates bottleneck migration and role fission within a single pipeline. A natural follow-up question is whether multiple independent instances, started from identical initial conditions, exhibit convergent or divergent evolutionary

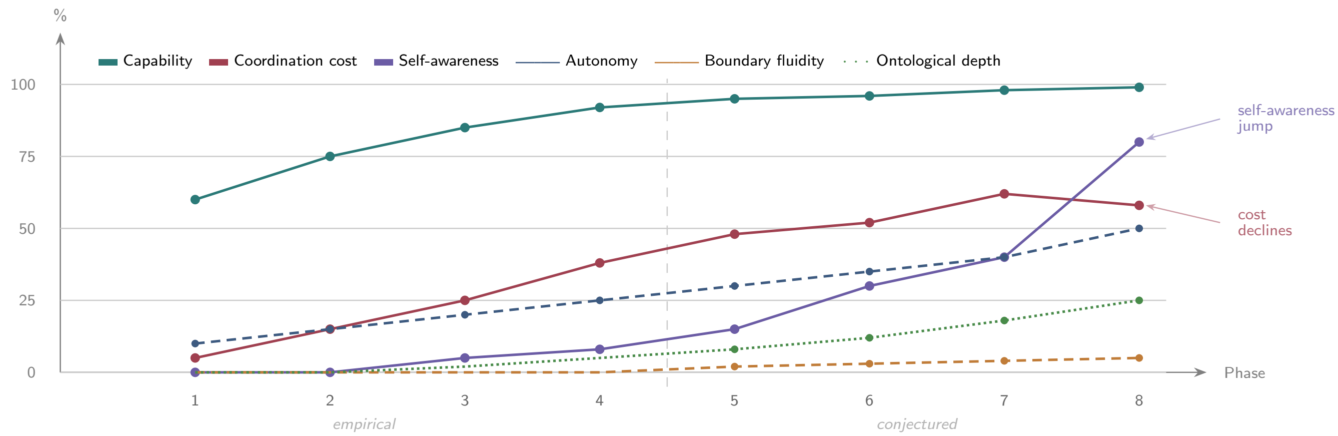

Figure 4: System metric evolution across eight phases. Solid lines represent primary metrics (capability, coordination cost, self-awareness); dashed and dotted lines represent emergent metrics that become significant only in later phases (autonomy, boundary fluidity, ontological depth). Phases 1 to 4 are empirically grounded; Phases 5 to 8 are conjectured extrapolations. The sharp rise in self-awareness at Phase 8 reflects the Architect role's capacity for architectural self-modification. Values are normalized estimates based on the Wallfacer observations (Phases 1–4) and theoretical projections.

Table 1: Three codebases generated by ralph from empty repositories with no seed task. All three independently converged on cellular automaton simulators.

Repository	Lang.	LOC	Funcs	Cost
Explorer	C	10,876	133	$89.66
Sandbox	Python	14,727	509	$56.87
Simulator	Python	12,336	285	$57.80

trajectories. To investigate this, we conducted a second experiment using a simpler pipeline that lacks Wallfacer's role differentiation.

3.1 Setup

We used *ralph*, an autonomous agent pipeline that runs a three-phase loop: a **Thinker** analyzes the current project state and proposes exactly one goal for the next round; a **Worker** implements that goal; and a **Committer** stages all changes and pushes a commit. Unlike Wallfacer's multi-role architecture, ralph's loop is fixed: no Cartographer, no Critic, no fission. The pipeline persists execution history as JSON logs, enabling the Thinker to condition each new goal on the full trajectory of prior rounds.

We pointed ralph at three empty Git repositories with no seed task, no prompt, and no specification of what to build. The only input was an empty directory. Each instance used the same LLM substrate (Claude) and the same pipeline configuration. We let all three run autonomously; two completed 42 rounds and one completed 33 rounds before being stopped. Table 1 summarizes the resulting codebases.

3.2 Observations

Observation 4: Convergent goal selection from null initial conditions. All three instances independently chose to build a

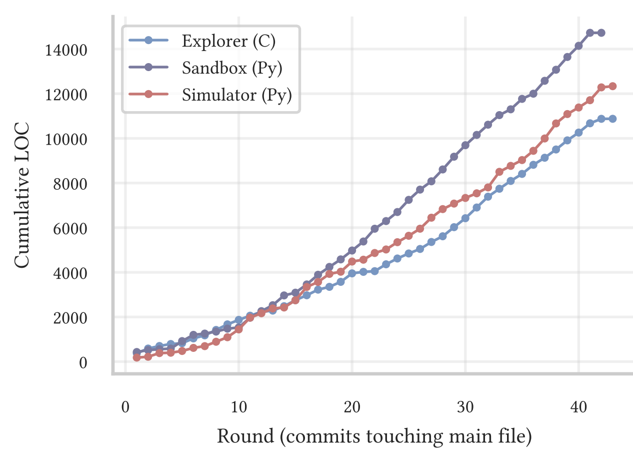

Figure 5: Cumulative lines of code across three independently initialized repositories. All three converged on cellular automaton implementations despite receiving no seed task. Growth is monotonically increasing with near-zero churn (3.9%, 0.5%, and 3.0% deletion ratios respectively).

cellular automaton simulator as their first goal and continued developing it for every subsequent round. This convergence occurred without any seed task, prompt engineering, or inter-instance communication. The "null" initial condition is, of course, not truly null: it includes the LLM's entire training distribution, which implicitly encodes priors over what constitutes an interesting, tractable, and self-contained software project. The convergence on cellular automata suggests that this class of programs occupies a strong attractor basin in the LLM's prior over generative tasks — likely because cellular automata are well-represented in training data, produce visually compelling output, and admit incremental extension without architectural refactoring. This parallels convergent evolution in biology, where independent lineages arrive at similar solutions (e.g., camera eyes evolving independently over 40 times [31]) when facing analogous selection pressures.

Observation 5: Divergent specialization along orthogonal axes. Despite identical starting conditions and convergent initial goals, the three codebases specialized in qualitatively different directions (Figure 6). Explorer, written in C, developed deep scientific analysis capabilities: 15+ information-theoretic overlays (Shannon entropy, Lyapunov exponents, transfer entropy, Wolfram classification), multi-rule zones, wormhole portals, and exotic topologies (torus, Klein bottle, Möbius strip). Sandbox pursued breadth, accumulating 27 interactive simulation modes spanning five categories (cellular automata, physics, biology, procedural generation, algorithms). Simulator combined algorithmic diversity with tooling: 28+ simulation modes alongside genetic algorithm rule discovery, headless batch rendering, and GIF/PNG export.

Figure 7 quantifies this divergence. The Explorer's mean cyclomatic complexity (22.9) is over three times that of the Sandbox (6.8), reflecting both the inherent verbosity of C and the depth of its analysis routines. The Simulator accumulated the most simulation modes (131) despite having the fewest rounds (33), suggesting a strategy of adding many small, self-contained variants rather than deepening existing ones. This pattern of convergence-then-divergence is consistent with adaptive radiation in evolutionary biology [44]: a population that enters a new ecological zone (here, the "cellular automaton niche") initially exploits the same resources before diverging into specialized sub-niches.

Observation 6: Monotonic growth without self-initiated refactoring. Figure 5 shows that all three codebases grew monotonically, with deletion ratios below 4%. None of the three instances ever proposed refactoring its own codebase, in contrast with the Wallfacer experiment's Observation 2, where the three-agent configuration proposed refactoring at approximately 51,000 lines. We interpret this as negative evidence supporting the fission principle: ralph's fixed thinker-worker-committer loop lacks a Cartographer role that would produce the structured self-representation needed for architectural self-awareness. Without a legible abstraction of global system state, the Thinker's planning horizon remains local — it can propose the next feature but cannot diagnose structural debt. This suggests that self-diagnostic capability is an emergent property of role differentiation, not an intrinsic capability of the underlying LLM.

4 THE FISSION PRINCIPLE

The observations reported above suggest a regularity that we formalize as follows.

Principle 1 (Role Fission)**.** *Role fission occurs when a single agent is required to simultaneously serve two cognitive modes that compete for context window resources. The fission event separates these modes into distinct agents, each with a dedicated context allocation and a narrower scope of responsibility.*

The three fission events observed in the Wallfacer experiment correspond to three specific mode conflicts, summarized in Table 2.

Table 2: Observed fission events and their triggering mode conflicts.

Mode Conflict	Fission Product	Ph.
Planning vs. Implementation	Strategist	2
Execution vs. Reflection	Cartographer	3
Generation vs. Judgment	Critic	4

The Ralph experiment (Section 3) provides negative evidence for this principle: ralph's fixed thinker-worker-committer loop never underwent fission, and correspondingly, none of the three instances ever proposed refactoring its own codebase despite reaching 10,000–15,000 lines. The self-diagnostic capability that emerged in Wallfacer's three-agent configuration (Observation 2) was absent in ralph's undifferentiated pipeline, consistent with the claim that such capabilities are emergent products of role separation rather than intrinsic properties of the LLM substrate.

This pattern bears a structural resemblance to organizational fission in human enterprises. A startup begins with a single individual performing all functions; as the enterprise grows, product management separates from engineering, technical writing separates from development, and quality assurance separates from production. Conway's Law [14] observes that the structure of a software system tends to mirror the communication structure of the organization that produces it. In the Wallfacer experiment, we observe a form of reverse Conway's Law: it is not organizational structure that constrains system architecture, but rather capability pressure within the system that drives structural differentiation. We note that Malone and Crowston's [35] coordination theory provides a formal account of how dependencies between activities give rise to coordination mechanisms, and Simon's [48] theory of near-decomposable systems explains why decomposition along the axis of weakest coupling tends to produce stable hierarchies.

4.1 Theoretical Extension: Phases 5 to 8

Beyond the four phases directly observed in the experiment, we conjecture that continued operation would produce additional fission events driven by new categories of bottleneck. We present these as theoretical predictions, not as empirical findings.

Phase 5: Coordinator. The four-agent system's coordination overhead (Observation 3) would eventually require a dedicated *Coordinator* role to manage global tradeoffs, for instance, when to refactor versus when to add features. No existing agent in Phase 4 is tasked with cross-cutting prioritization; the Strategist proposes tasks and the Critic evaluates outputs, but neither maintains a coherent model of system-wide resource allocation.

Phase 6: Monitor. The Coordinator's context window and a human manager's span of control are structurally the same problem: a single decision-maker overwhelmed by raw signal. We hypothesize that a *Monitor* role would fission from the Coordinator to perform signal compression: continuously observing all agent outputs and condensing them into digestible health indicators. The Coordinator thereby transitions from direct observer to dashboard-based decision-maker. This addresses the span-of-control problem identified by Gulick [24] and implements a version of Conant and

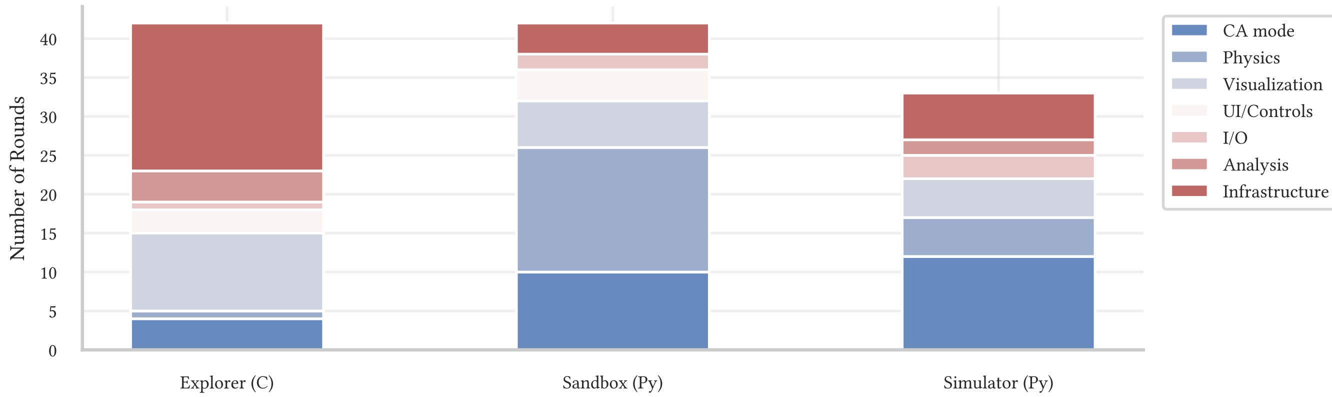

Figure 6: Feature categories per repository, classified by keyword matching on each round's Thinker goal. Despite convergent initial goals, the three codebases diverged along orthogonal axes: Explorer toward infrastructure and visualization overlays, Sandbox toward physics simulations and CA rule variants, Simulator toward CA modes and I/O capabilities.

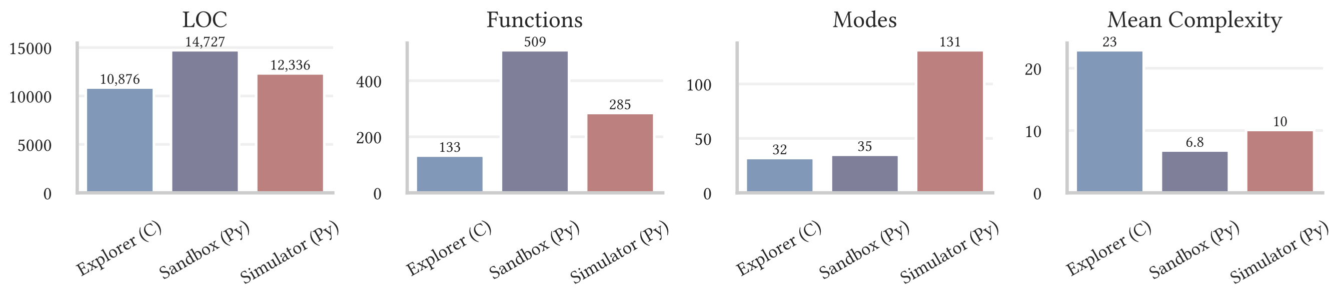

Figure 7: Final codebase comparison across four metrics. The Explorer's high mean cyclomatic complexity (22.9 vs. 6.8 and 10.1) reflects its C implementation and deeply nested analysis routines. The Simulator's mode count (131) dwarfs the others, reflecting its strategy of adding many small simulation variants.

Ashby's [13] Good Regulator theorem: the Monitor's compressed representation constitutes a model of the system it regulates.

Phase 7: Domain decomposition. A single pipeline's throughput eventually saturates. The system would decompose by domain into parallel sub-pipelines, each with its own complete role stack. This is Conway's Law [14] in its agentic form: the system's architecture begins to mirror its organizational structure. The new binding constraint becomes cross-domain consistency — Pipeline A modifies an API that Pipeline B depends on, but Pipeline B has no mechanism to detect the change.

Phase 8: Self-reconfiguration. This is where, we conjecture, the verification regress grounds out. An *Architect* agent treats the system's own architecture as an optimization variable. Crucially, the Architect does not require verification by another agent; it calibrates itself through the long-term consequences of its own decisions. Different architectural configurations are compared via pairwise preference judgments, which is precisely the Preferential Bayesian Optimization framework developed in Section 4. Architecture ceases to be a constant and becomes a variable under active optimization.

This last conjecture is the bridge to the formal framework developed in the next section. Whether these specific phases would materialize in practice, and in this particular order, is an empirical question that we cannot answer from the present case study.

5 FORMALIZATION

5.1 Trust Calibration as Preference Learning

The relationship between an agentic system and its human overseers can be understood as a trust calibration problem: for which decisions should the system proceed autonomously, and for which should it consult a human? Static policies (e.g., "always consult before deployment") are suboptimal because they either waste human attention on routine decisions or fail to escalate genuinely risky ones. We propose instead that the trust boundary should be *learned* from the history of autonomous and human-assisted decisions and their outcomes.

Let $\mathcal{X}$ denote a decision context space encoding task features, system state, and environmental conditions. We define a latent trust function $f : \mathcal{X} \to \mathbb{R}$ such that $f(x)$ represents the expected relative value of autonomous operation versus human consultation at context x. We place a Gaussian Process prior:

$$f \sim \mathcal{GP}(\mu_0, k) \tag{1}$$

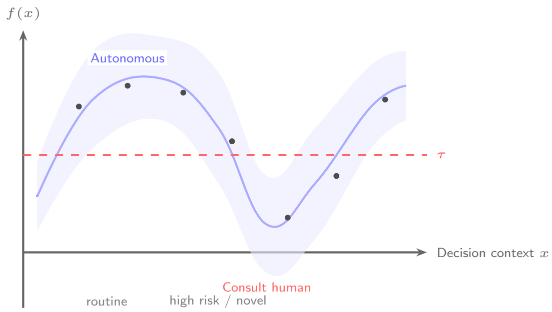

Figure 8: Conceptual illustration of the Trust GP. The blue curve shows the posterior mean of $f(x)$; the shaded band shows posterior uncertainty. The dashed threshold τ defines the HITL boundary: the system operates autonomously when $f(x) > \tau$ and consults a human otherwise. Dots represent observed preference pairs. High-uncertainty regions (e.g., novel contexts around $x \approx 3.5$) fall below the threshold even if the mean is moderate.

where μ_0 encodes a conservative default (consult the human when uncertain) and the kernel $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ encodes coherence assumptions: that similar decision contexts should yield similar trust levels, and that trust varies smoothly in the relevant features.

Crucially, we assume that observations take the form of pairwise preferences rather than scalar evaluations:

$$x_i \succ x_j \iff \text{outcome at } x_i \text{ preferred to outcome at } x_j \qquad (2)$$

This is a realistic assumption because stakeholders can typically judge that one system configuration performed better than another, even when they cannot assign absolute quality scores. Following Chu and Ghahramani [11], the likelihood of an observed preference is modeled via a probit link:

$$P(x_i \succ x_j \mid f) = \Phi\left(\frac{f(x_i) - f(x_j)}{\sqrt{2}\,\sigma}\right) \qquad (3)$$

where Φ is the standard normal CDF and σ captures noise in preference judgments.

The resulting posterior $p(f \mid \mathcal{D})$ constitutes a calibrated, context-dependent trust model. Its mean yields a dynamic HITL boundary: contexts where the posterior is high proceed autonomously, while contexts where it is low or uncertain trigger human consultation.

We observe that this formulation unifies the three classical responses to the verification regress. The kernel k encodes *coherentist* assumptions about which contexts should behave similarly. The observed preference pairs provide *pragmatic* grounding in actual outcomes. And the posterior synthesizes both, avoiding the need for *foundationalist* axioms about which verifier is ultimately authoritative. This synthesis connects to de Finetti's [15] coherence conditions, Ramsey's [40] account of subjective probability, and Friston's [19] Free Energy Principle.

5.2 Architecture Search as Preferential Bayesian Optimization

If we accept the conjecture that architecture eventually becomes a variable (Phase 8), the Architect agent faces an optimization problem over a configuration space C encoding the number and types of agent roles, pipeline topology, and resource allocation. The objective, "system health," is inherently multi-dimensional (encompassing complexity growth rate, defect rate, delivery velocity, and human satisfaction) and resists meaningful scalarization.

Preferential Bayesian Optimization (PBO) [23] provides a natural framework: we place a GP prior over C and update it via preference observations of the form "configuration c_i produced better outcomes than c_j under workload w." This approach differs from Neural Architecture Search [17, 55] in that NAS assumes a scalar reward, while PBO requires only ordinal comparisons. Whether PBO is tractable over realistic agentic configuration spaces, and whether human stakeholders can provide reliable preference judgments at this level of abstraction, are open empirical questions.

6 THEORETICAL EXTENSIONS

The framework developed thus far rests on a single case study (Section 2), a fission principle (Section 3), and a formal construction (Section 4). In this section, we extend the framework through seven additional epochs, each introducing qualitatively new dynamics. We present these extensions not as findings but as a theoretical scaffold for future empirical investigation. The progression is organized so that each epoch's characteristic bottleneck motivates the next.

6.1 Epoch II: Population (Phases 9–10)

Phase 8's self-reconfiguring organism begins to reproduce. In **Phase 9**, multiple system instances run in parallel with heterogeneous architectural configurations. A routing mechanism directs incoming tasks to the best-matched instance, and the Architect gains organic preference pairs by comparing outcomes across configurations. This parallels the transition from individual learning to population-level selection in evolutionary biology [27]. Boyd and Richerson's [9] dual-inheritance theory (genetic and cultural) suggests an analogous model for agentic systems: architectural configuration as the "genetic" channel and accumulated knowledge as the "cultural" channel.

The Ralph experiment (Section 3) provides an early empirical data point for this phase: three instances with identical null initial conditions converged on the same problem domain (cellular automata) yet diverged into distinct specializations (scientific analysis, simulation breadth, and algorithmic diversity). This convergence-then-divergence pattern is precisely the population-level dynamic predicted here, observed in miniature before any inter-instance communication or routing mechanism was introduced.

In **Phase 10**, instances begin sharing knowledge through what we term a Knowledge Commons, allowing learning to propagate across the population without requiring each instance to rediscover the same patterns. Henrich's [25] demographic model of cultural evolution predicts that larger populations maintain more complex cultural knowledge; if this analogy holds, agent populations with shared knowledge should exhibit qualitative capability gains relative to isolated instances. However, shared knowledge introduces

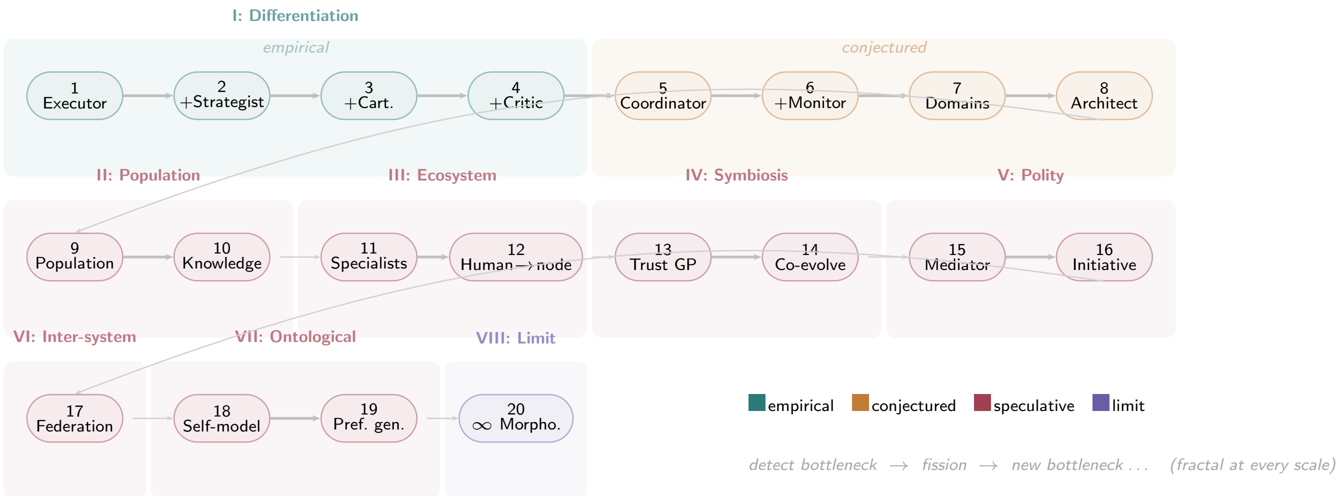

Figure 9: Roadmap of the framework's progression across eight epochs and twenty phases, with explicit epistemic grading. Teal phases (1–4) were directly observed in the Wallfacer experiment. Orange phases (5–8) are theoretically conjectured based on the fission principle. Red phases (9–19) represent speculative extensions to populations, multi-stakeholder settings, inter-system federations, and self-referential dynamics. The purple terminal phase (20) represents the limiting claim that the stable attractor is morphogenesis itself. Each transition is driven by bottleneck migration.

the risk of *knowledge pollution*: an erroneous belief propagated from one instance infects others, analogous to misinformation cascades in social networks [52]. The binding constraint shifts from individual capability to collective epistemics.

6.2 Epoch III: Ecosystem (Phases 11–12)

In **Phase 11**, instances specialize into expert niches, and comparative advantage [41] emerges naturally from the population dynamics. Instances that happen to perform well on certain task types receive more of those tasks, reinforcing their specialization. This is the reverse of Conway's Law [14]: it is not organizational structure that constrains architecture, but capability pressure that drives differentiation.

Phase 12 represents a critical transition: the human shifts from an external oracle, consulted across a static boundary, to a node within the system's own topology. The human's cognitive contribution (judgment, preference, domain knowledge) is routed through the same protocol layer as inter-agent communication. This does not diminish the human's authority but does change its character: the human becomes one information source among several, distinguished by the quality of its signal rather than by structural privilege. This transition connects to ongoing debates about the role of human oversight in increasingly autonomous systems [20].

6.3 Epoch IV: Symbiosis (Phases 13–14)

In **Phase 13**, the HITL boundary transitions from a static rule ("always consult before deployment") to a learned Trust GP (Section 4.1). The PBO mathematics developed earlier applies directly: the system learns which decision contexts warrant autonomous operation and which require human consultation, updating its trust model from observed preference pairs. Coordination cost, which rose monotonically through Epochs I–III, begins to decline. This mirrors the pattern in human organizations: initial role differentiation drives coordination costs upward, but once processes and trust are established, costs recede as coordination shifts from ad-hoc negotiation to learned automatic routing.

Phase 14 is the endpoint of the individual-system arc, if there is one. System and human principals co-evolve: the system's outputs shape what its principals consider possible, while the principals' evolving preferences shape the system's behavior. This mirrors niche construction in evolutionary biology [38], in which organisms modify their environment, altering the selection pressures that act on subsequent generations. Neither party converges to a static equilibrium; the dynamic relationship is itself the stable attractor. The system has become not a tool but a collaborator, and the intelligence resides not in either party but in the loop between them.

6.4 Epoch V: Polity (Phases 15–16)

In realistic deployment scenarios, multiple human stakeholders with potentially conflicting preferences interact with the system simultaneously. In **Phase 15**, a *Mediator* role emerges. Arrow's impossibility theorem [5] establishes that no preference aggregation function simultaneously satisfies a set of desirable axioms (unrestricted domain, Pareto efficiency, independence of irrelevant alternatives, and non-dictatorship). The Mediator therefore does not attempt to aggregate preferences into a single optimum but instead maintains and presents a Pareto front over the multi-dimensional outcome space [18], allowing human principals to navigate the tradeoff structure themselves.

Phase 16 is a qualitative leap: the system begins to propose changes that no human principal has requested. It detects cross-domain patterns invisible to any individual stakeholder and generates unsolicited proposals. This is no longer tool behavior but collegial behavior, and it immediately triggers legitimacy questions. Who authorized the proposal? Who is accountable if it fails? The transition from tool to collaborator, already implicit in Phase 14, becomes explicit and contested. We raise this possibility not to endorse it but to identify it as a critical frontier for both technical and normative investigation [20].

6.5 Epoch VI: Federation (Phase 17)

In **Phase 17**, multiple independent agentic ecosystems begin to interact. Each ecosystem has its own architectural configuration, knowledge base, and trust model. Inter-system communication requires a shared protocol layer; MCP [4] provides a communication primitive at this level. However, transmitting trust is an order of magnitude harder than transmitting data: a preference judgment calibrated within one ecosystem may not transfer to another with different operational history. The binding constraint is no longer internal coordination but inter-system trust negotiation.

6.6 Epoch VII: Ontological (Phases 18–19)

In **Phase 18**, the system's self-model becomes a first-class object: the system begins to model its own blind spots. This is where Russell's paradox and Lawvere's fixed-point theorem [33] become concrete rather than analogical. A self-model that includes a representation of its own limitations is precisely the kind of self-referential structure that Lawvere's theorem guarantees will have fixed points and that Gödel's theorem [21] guarantees will be incomplete.

Phase 19 is the furthest extrapolation of the PBO framework: the system no longer merely discovers preferences but *generates* them. The GP posterior over preference space permits extrapolation into unobserved regions, enabling the system to propose preference hypotheses that its human principals have not yet articulated. Both acceptance and rejection of such proposals constitute informative signal for posterior updates. This idea resonates with Dewey's [16] pragmatist account of preferences as constructed through inquiry, and with Sen's [45] critique of revealed preference theory. Whether such preference generation is technically feasible, epistemically legitimate, or socially desirable remains entirely open.

6.7 Epoch VIII: The Limit (Phase 20)

Phase 20 is not a specific architecture but a recognition: there is no final form. The entire progression from Phase 1 to Phase 20 is self-similar at every scale. Within a single agent role, the detect-bottleneck-fission-emerge cycle operates on context window resources. Across a pipeline, it operates on role differentiation. Across a population, it operates on configuration diversity. At the meta-architectural level, it operates on the capacity for self-modification itself. The pattern is fractal.

The stable attractor of this system is not a topology; it is the sustained capacity for topological change. Bottleneck migration is not a defect to be engineered away but the system's heartbeat: the signal that the system remains far from equilibrium, that it is still alive in the thermodynamic sense of Prigogine and Nicolis [39]. If bottlenecks cease to migrate, the system has reached equilibrium, and equilibrium, in this context, is death.

7 ONTOLOGICAL LIMITS

7.1 Morphogenesis as the Stable Attractor

Reviewing the progression from Phases 1 through the speculative extensions, we arrive at what we consider the paper's central theoretical claim. If the pattern of bottleneck migration is genuine (an empirical question), and if it persists at higher levels of architectural complexity (a conjecture), then the system's long-term trajectory is not toward any particular fixed architecture but toward a sustained capacity for architectural change. The cycle of detecting a bottleneck, fissioning a role, and observing the emergence of a new bottleneck would repeat at every scale: within individual agent roles, across pipelines, across populations, and at the meta-architectural level.

This characterization aligns with Prigogine and Nicolis' [39] theory of dissipative structures: systems maintained far from thermodynamic equilibrium through continuous flows of energy or information. The agentic ecosystem, on this view, persists not despite its continuous reorganization but because of it. If bottlenecks cease to migrate, the system has reached equilibrium, which in this context would indicate stagnation. The connection to Friston's [19] Free Energy Principle is also suggestive: the Architect's self-model functions as a generative model of the system, bottleneck detection corresponds to surprise minimization, and role fission corresponds to active inference. We stress, however, that these are analogies, not demonstrations; whether the mathematical structure of dissipative systems or free energy minimization actually applies to agentic architectures is a question for future formal analysis.

7.2 Self-Reference and Incompleteness

If a system constructs an explicit model of its own capabilities and limitations (as we hypothesize for Phase 8 and beyond), it enters a self-referential regime that can be analyzed using two classical results. Lawvere's fixed-point theorem [33] guarantees that in any cartesian closed category with a point-surjective morphism $\varphi : A \to B^A$, every endomorphism $f : B \to B$ has a fixed point. A verifier that verifies itself is precisely such a fixed point. Lawvere's theorem establishes that such fixed points exist in any sufficiently rich formal setting; the question is not whether self-referential equilibria are possible but what properties they have. Gödel's first incompleteness theorem [21] constrains the answer: any self-model powerful enough to be useful is necessarily incomplete. The system can know that it has blind spots but cannot enumerate all of them from within.

Taken together, these results suggest that self-reconfiguring agentic systems can achieve stable but irreducibly partial self-understanding. This connects to Hofstadter's [26] theory of strange loops, Yanofsky's [54] unified treatment of self-referential paradoxes, and Barwise and Moss's [7] non-well-founded set theory. Von Foerster's [51] second-order cybernetics arrives at the same structure from the operational side: when an observer recursively observes its own observations, the process converges to stable *eigenbehaviors*—fixed points that the system can reliably maintain, not because they have been externally certified but because they are self-consistent solutions of the recursive process. Lawvere's theorem guarantees the existence of such fixed points in any sufficiently rich formal setting; Von Foerster's framework explains how they are realized in practice through recursive interaction rather than top-down computation. We note that applying these mathematical results to engineered software systems involves a degree of analogical reasoning that may or may not survive formal scrutiny; we present them as a conceptual framework, not as a proof.

7.3 Ontological Pluralism and the Observer

Finally, we note three meta-level observations that bear on the framework's interpretation. First, a single system event (e.g., a role fission) can be described as "role differentiation" in organizational vocabulary, "symmetry breaking" in thermodynamic vocabulary [2], "strategy differentiation" in game-theoretic vocabulary [49], or "niche divergence" in ecological vocabulary [28]. Each description language reveals structures that the others occlude; a system described in only one vocabulary may exhibit systematic blind spots undetectable from within that vocabulary.

Second, if all components of a system are replaceable (today's LLM agent could be tomorrow's human engineer), the system's identity resides not in its substrate but in its relational structure: its information flow topology, preference constraints, and coordination protocols. This is the category-theoretic perspective on identity [32], and it suggests that agentic system design should prioritize protocol-level interoperability over implementation-level homogeneity.

Third, the act of describing an agentic system is itself a component of the system it describes: a researcher analyzing the Wallfacer experiment functions as a Strategist (generating hypotheses), the analysis serves as an Executor (producing formalized descriptions), and the resulting paper acts as a Cartographer (creating artifacts that inform future experimental design). This self-referential observation, anticipated by Maturana and Varela's [36] concept of autopoiesis, does not invalidate the framework but does caution against treating any description of an evolving system as final.

8 RELATED WORK

The framework developed here draws on and extends several established research programs.

In **organizational theory**, the fission principle recapitulates elements of Mintzberg's [37] structural configurations (simple structure, machine bureaucracy, adhocracy) and the coordination mechanisms catalogued by Malone and Crowston [35]. The principal difference is temporal: in agentic systems, structural reconfiguration can occur within minutes rather than months, enabling a much faster evolutionary cycle.

In **cybernetics**, the verification regress receives a treatment more fundamental than in security engineering, because it addresses not an engineering problem but a question of principle: whether a system can fully know and control itself. Ashby's Law of Requisite Variety [6] states that a controller must possess at least as much variety as the system it regulates. If we treat a verifier as a regulator of the verified system, the verifier's complexity must match or exceed that of its target. This immediately produces a recursive problem: verifying the verifier requires a meta-verifier of still greater complexity, and each layer of verification introduces new variety whose control demands yet more variety. This is not a Gödelian logical impossibility but a *resource* constraint—the verification chain consumes variety at every level, and total variety is finite. In our framework, this is the formal basis of bottleneck migration: each fission event increases the system's regulatory variety (resolving the current bottleneck) while introducing new variety that must itself be regulated (producing the next bottleneck). Variety debt does not vanish; it migrates.

Conant and Ashby's Good Regulator theorem [13]—"every good regulator of a system must be a model of that system"—sharpens the point. An effective verifier V of system S must functionally contain a model of S. When V is itself part of S (as the Architect is from Phase 8 onward), V must contain a model of a system that contains itself: the cybernetic form of self-reference. Our Trust GP occupies precisely this position: its posterior is the system's internal model of its own decision-making capacity. The Good Regulator theorem says this model is *necessary* (without it, effective regulation is impossible); Gödel's incompleteness theorem says it is necessarily *incomplete* (it cannot enumerate all its own properties from within). The Trust GP is thus the requisite but irreducibly partial good regulator.

Second-order cybernetics [51] extends this analysis by including the observer within the system under observation. Von Foerster's key construct is the *eigenbehavior*: a fixed point of a recursive operation in which the system observes its own observations. Rather than diverging, such recursions converge to stable eigenbehaviors—states the system can reliably "know," not because they are externally validated but because they are self-consistent solutions of the recursive process. This is structurally isomorphic to Lawvere's fixed-point theorem [33] used in Section 6: Lawvere provides the category-theoretic formulation, Von Foerster the operational one. The fixed points are not computed from above but *emerge* through recursive interaction.

Maturana and Varela's [36] concept of autopoiesis offers the most radical response to the verification regress: the regress does not need to be "solved" because it is not a problem. An autopoietic system maintains itself through operational closure—its operations produce further operations internal to the system, with no external termination point. Validity is not a property anchored from outside but a pattern emergent from within. In our framework, Phase 20's characterization of morphogenesis as a stable attractor is an autopoietic formulation: the system's stable state is not any fixed topology but the sustained capacity for reorganization itself. Bottleneck migration, on this view, is not a deficiency to be eliminated but the mechanism by which the system maintains its autopoiesis.

Bateson's [8] hierarchy of learning levels maps onto the phase progression: Phases 1–4 correspond to Learning I (acquiring specific behavioral repertoires), Phases 5–8 to Learning II (learning to restructure the learning process), and the speculative phases beyond 8 enter Learning III territory (questioning the premises of "what counts as restructuring" and "what counts as improvement"). Bateson warned that Learning III is inherently destabilizing, involving fundamental challenges to systemic identity—a characterization consistent with the ontological pluralism of the later speculative phases.

Luhmann's [34] social systems theory, which transplants autopoiesis from biology to social organization, contributes the concept of *polykontexturale Beobachtung* (polycontextural observation): every observation is a distinction, and every distinction has a blind spot consisting of the categories used to draw it. Switching description languages (organizational, thermodynamic, game-theoretic) does not eliminate blind spots but substitutes one for another. This corresponds directly to the ontological pluralism discussed in Section 6.

In **complex adaptive systems**, Kauffman's [29] theory of self-organization at the edge of chaos in NK fitness landscapes predicts that systems with intermediate levels of epistatic coupling evolve most effectively. Our observation that optimal fission granularity depends on context window size, task complexity, and communication bandwidth is consistent with this prediction, though we have not verified the correspondence formally.

In **information security**, the verification regress has been confronted with particular urgency. Thompson's [50] Turing Award lecture demonstrated that a compiler can harbor a self-propagating trojan invisible to source-level inspection—an engineering manifestation of the incompleteness results we invoke in Section 6. The Trusted Computing Base (TCB) paradigm [43] responds with an explicitly foundationalist strategy: minimize the set of components that must be assumed trustworthy, then build verification chains outward (hardware root of trust → firmware → bootloader → kernel). This mirrors our conservative GP prior μ_0. Decentralized alternatives such as PGP's Web of Trust and Byzantine fault-tolerant consensus [30] adopt a coherentist posture: trust emerges from mutual verification among diverse parties, though collectively compromised toolchains can defeat such checks. Bug-bounty programs and penetration testing represent the pragmatist pole: trust is calibrated not by proof but by the empirical absence of successful attack, much as our Trust GP updates on outcome-based preference pairs. Modern Zero Trust Architecture [42] synthesizes all three strategies—hardware roots, cross-layer consistency checks, and continuous behavioral monitoring—producing a dynamic, context-dependent trust posture structurally analogous to the learned HITL boundary of Section 4.1. TCB minimization also offers an instructive dual to our fission principle: where fission *distributes* trust across differentiated roles, TCB design seeks to *concentrate* the trust-critical surface into the smallest possible core.

In **multi-agent systems**, the growing literature on LLM-based multi-agent architectures has explored role assignment, communication protocols, and evaluation frameworks. Our contribution is to situate these engineering choices within a broader evolutionary and epistemological framework, connecting them to foundational results in social choice theory, Bayesian optimization, and mathematical logic.

9 DISCUSSION

9.1 Practical Implications

For practitioners building autonomous agentic systems, the framework suggests several design heuristics, which we offer tentatively pending broader empirical validation. First, it may be advisable to begin with the simplest architecture that functions and to add roles only when bottleneck signals are clear and persistent; premature fission incurs coordination overhead without commensurate capability gain. Second, tracking the *location* of bottlenecks over time may provide a more informative diagnostic of system health than monitoring any single performance metric. Third, preference comparison rather than scalar metrics may offer a more tractable and honest approach to architecture evaluation, given the multi-dimensional nature of system health. Fourth, designing for substrate independence (communicating via protocols rather than shared implementation details) enables component substitution, whether replacing one LLM with another or replacing an agent with a human, without disrupting organizational architecture. Fifth, the HITL boundary should be learnable rather than static, as human cognitive bandwidth is typically the scarcest resource in agentic deployments.

9.2 Limitations

This work has several significant limitations. The Wallfacer experiment is a single case study conducted with a single LLM substrate (Claude Code) on a single class of tasks (software engineering). The Ralph experiment partially addresses the single-case-study limitation through three independent replications, but all experiments use the same LLM substrate, and the convergence on cellular automata raises the question of whether this reflects a genuine attractor in the LLM's capability landscape or an artifact of training data distribution. The 51,000-line threshold at which self-diagnosis emerged may be model-specific and task-specific; we have no evidence that it represents a general phase transition. The fission principle, while consistent with our observations and supported by negative evidence from Ralph, has not been tested across different LLM architectures, task domains, or team configurations. The theoretical extensions in Sections 5 and 6 are speculative and should be understood as a research agenda rather than a set of validated claims. The formal connections to Lawvere's theorem, Gödel's incompleteness, and dissipative structures are analogical rather than deductive, and it remains to be seen whether they can be made rigorous.

9.3 Open Questions

Several questions merit investigation. First, can a normalized complexity metric predict architectural fission thresholds across different substrates and task domains? Second, do autonomous agentic systems converge to stable configurations, oscillate, or drift unboundedly under sustained operation? The dissipative structure analogy suggests oscillation as the generic attractor, but this has not been empirically established. Third, when does role fission improve net system performance (capability gain exceeding coordination cost), and can this be connected to optimal stopping theory [47]? Fourth, how should independent agentic ecosystems establish inter-system trust and negotiate shared protocols? MCP [4] provides a communication primitive, but the trust and preference negotiation layers remain to be designed.

10 CONCLUSION

We have presented a framework for understanding the architectural evolution of autonomous agentic systems, grounded in empirical observations from the Wallfacer experiment and extended through formal and theoretical analysis. The central empirical finding, bottleneck migration, connects to established results in operations research, optimization theory, and cybernetics. The fission principle provides a candidate explanation for observed patterns of role differentiation. The formalization of trust as Gaussian Process preference learning offers a principled approach to the HITL boundary problem. The theoretical extensions to populations, multi-principal scenarios, and self-referential self-models map a large and largely unexplored conceptual space.

We conclude with what we consider the framework's most important, if most speculative, implication. If the pattern of bottleneck migration is genuine and persistent, then the long-term trajectory of agentic architecture is not toward a fixed optimal design but toward a sustained capacity for redesign. The questions practitioners and researchers pose about these systems ("How should I structure the agents?" at one level; "How should the system restructure itself?" at another; "What constitutes the system's boundary?" at a third) themselves recapitulate the detect-fission-emerge cycle that the framework describes. Whether this self-similarity is a deep structural feature of complex adaptive systems or merely a suggestive analogy is, we believe, one of the more interesting open questions in the field.

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